



Independent University, Bangladesh
School of Engineering Technology and Science
Department of Computer Science and Engineering

CSE 426-PROJECT PROPOSAL FORM

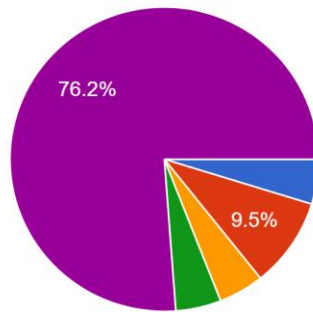
SEMESTER: Autumn 2026

PROJECT TITLE: Smart Irrigation Rover for Adaptive Spot Watering

Survey to develop a process for complex engineering problems considering cultural and societal factors:

Age Group(বয়সের শ্রেণি)

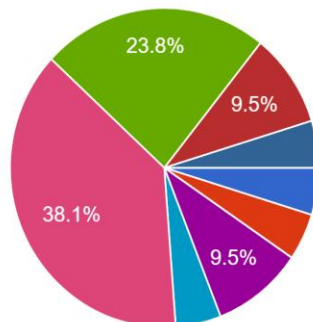
21 responses



- 20-30 (২০-৩০ বছর)
- 31-40 (৩১-৪০ বছর)
- 41-50 (৪১-৫০ বছর)
- Above 50 (৫০ বছরের বেশি)
- 20-30

Occupation (পেশা)

21 responses

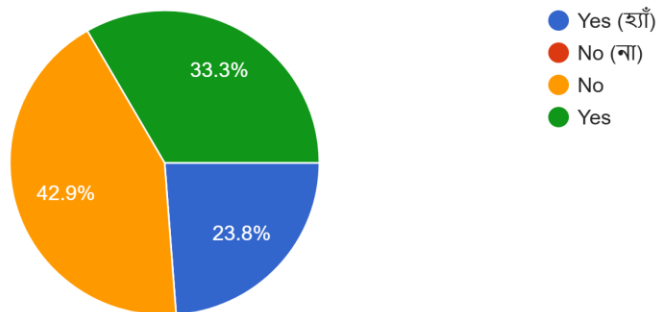


- Farmer (কৃষক)
- Gardener (বাগান পরিচর্যাকারী)
- Nursery Owner (নার্সারি মালিক)
- Business Owner (ব্যবসায়ী)
- Agriculture Owner (কৃষি খামারের মালিক)
- Other (অন্যান্য)
- Other
- Gardener

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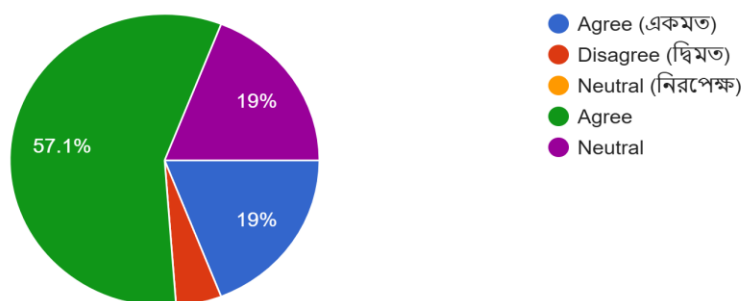
Are you involved in farming or plant cultivation? (আপনি কি কৃষিকাজ বা উদ্ভিদ চাষের সাথে জড়িত?)

21 responses



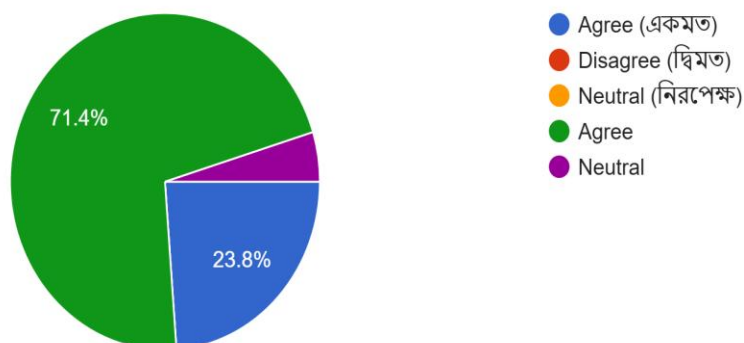
Do you think manual irrigation wastes water? (আপনি কি মনে করেন হাতে পরিচালিত সেচ পদ্ধতিতে পানির অপচয় হয়?)

21 responses



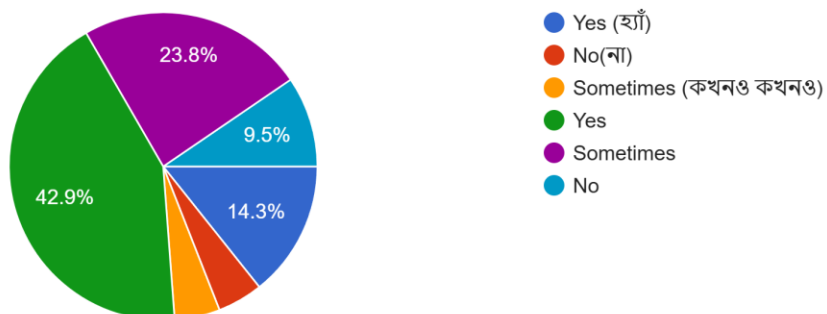
Is manual irrigation time-consuming? (হাতে পরিচালিত সেচ কি সময়সাপেক্ষ?)

21 responses



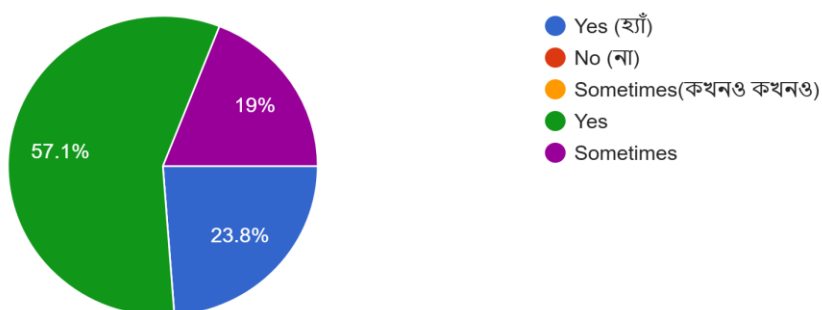
Do you face difficulties monitoring soil conditions regularly? (মাটির অবস্থা নিয়মিত পর্যবেক্ষণে কি আপনি সমস্যার সম্মুখীন হন?)

21 responses



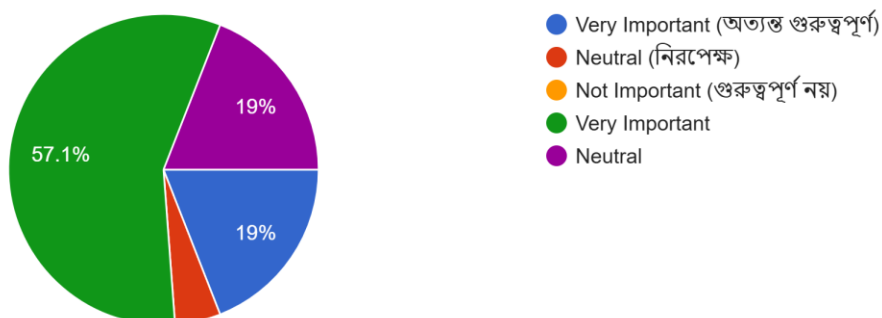
Do you think overwatering can damage crops or plants? (আপনি কি মনে করেন অতিরিক্ত পানি ফসল বা গাছের ক্ষতি করতে পারে?)

21 responses



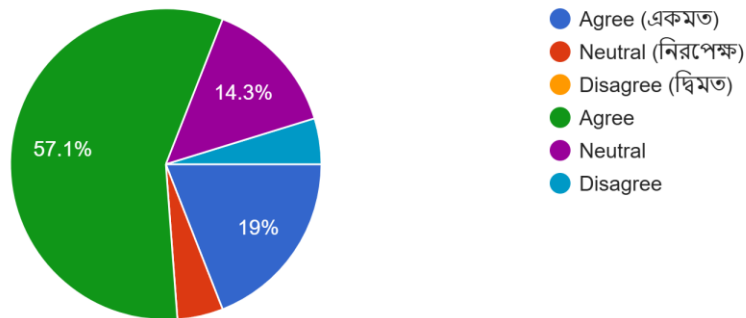
How important is soil moisture monitoring in agriculture? (কৃষিক্ষেত্রে মাটির আর্দ্রতা পর্যবেক্ষণ কতটা গুরুত্বপূর্ণ?)

21 responses



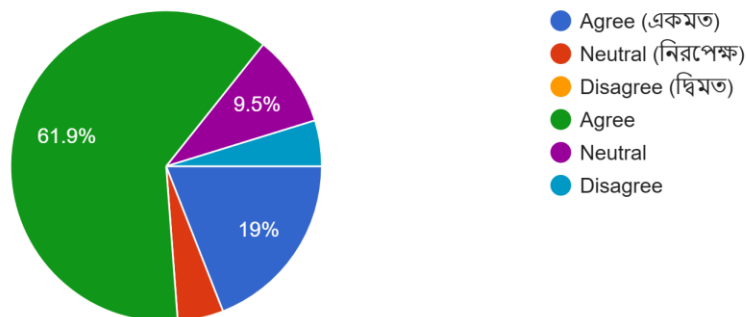
Would an automated irrigation robot help reduce labor? (একটি স্বয়ংক্রিয় সেচ রোবট কি শ্রম কমাতে সাহায্য করবে?)

21 responses



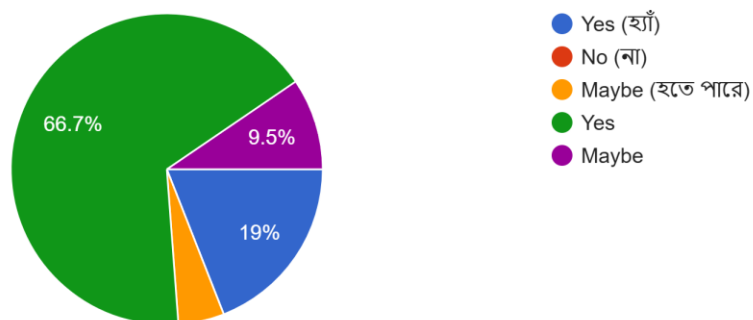
Do you think smart farming technology can improve agriculture? (আপনি কি মনে করেন স্মার্ট কৃষি প্রযুক্তি কৃষির উন্নতি করতে পারে?)

21 responses



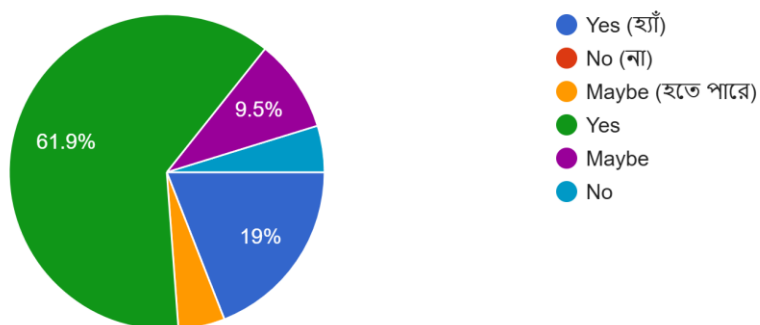
Would you like a system that can automatically detect dry soil and irrigate it?(আপনি কি এমন একটি ব্যবস্থা চান যা স্বয়ংক্রিয়ভাবে শুকনো মাটি শনাক্ত করে সেচ প্রদান করবে?)

21 responses



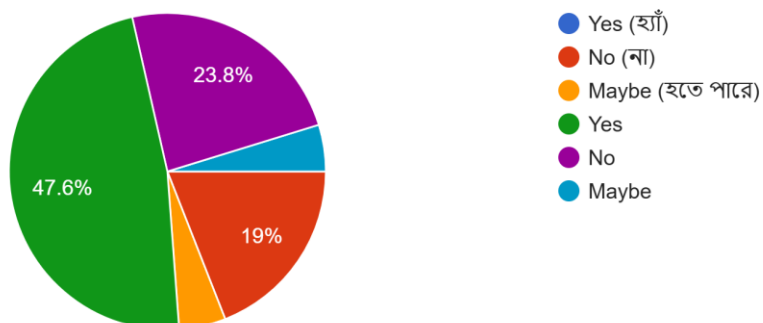
Is obstacle avoidance important for autonomous farming robots? (স্বয়ংক্রিয় কৃষি রোবটের জন্য বাধা শনাক্তকরণ ও এড়িয়ে চলার সুবিধা কি গুরুত্বপূর্ণ?)

21 responses



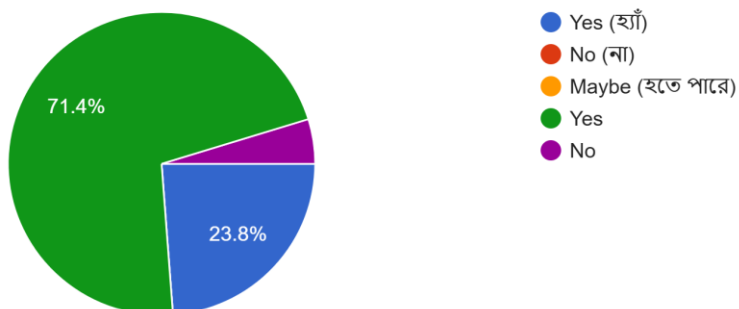
Have you heard about IoT-based smart farming before? (আপনি কি আগে IoT ভিত্তিক স্মার্ট কৃষি সম্পর্কে শুনেছেন?)

21 responses



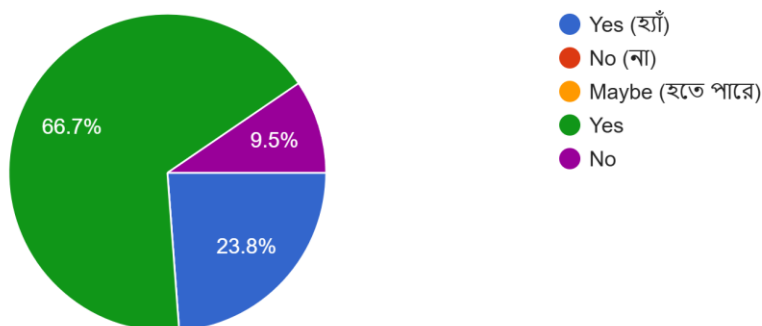
Do you believe AI and automation can improve future agriculture? (আপনি কি বিশ্বাস করেন AI ও স্বয়ংক্রিয় প্রযুক্তি ভবিষ্যতের কৃষিকে উন্নত করতে পারে?)

21 responses



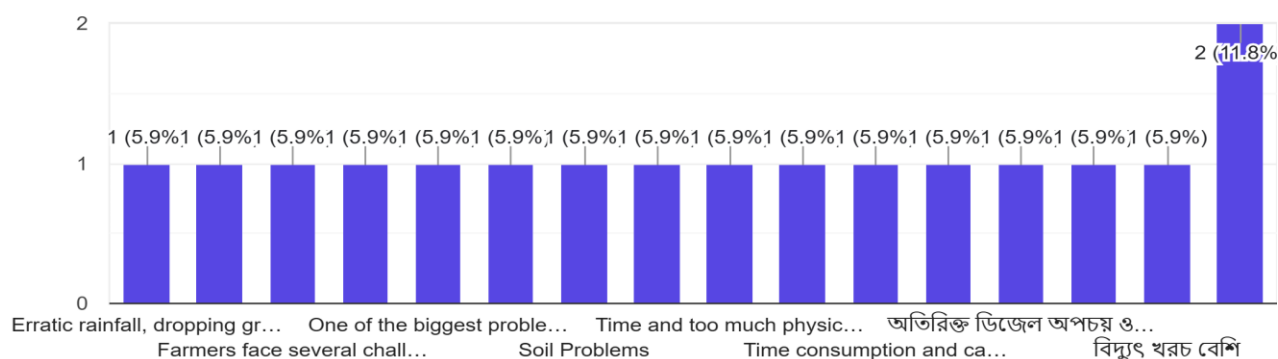
Would you be interested in using low-cost smart farming robots?(আপনি কি স্বল্পমূল্যের স্মার্ট কৃষি রোবট ব্যবহার করতে আগ্রহী?)

21 responses



What are the biggest problems farmers face during irrigation? (সেচ প্রদানের সময় কৃষকদের সবচেয়ে বড় সমস্যাগুলো কী?)

17 responses



What features would you expect from a smart farming robot?(একটি স্মার্ট কৃষি রোবট থেকে আপনি কী কী সুবিধা বা বৈশিষ্ট্য প্রত্যাশা করেন?)

16 responses

- crop monitoring, precision weeding, and harvesting using AI, sensors, and GPS navigation.
- Automatic irrigation, crop monitoring, pest detection, and real-time farm data.
- স্বল্প খরচে সেচ ব্যবস্থা, কম সময়ে আগাছা পরিষ্কার করা, ধান কাটা ও মাড়াই সম্পন্ন করা
- A smart farming robot should include automatic irrigation, sensors for soil and weather monitoring, crop and pest detection, GPS navigation, remote control, and efficient water, fertilizer, and energy management to make farming easier and more productive.
- Routine wise watering, detection of dehydration of plants, proper soil quality monitoring
- Perfect irrigation system
- সময়উপযোগি সকল কাজ অতি দ্রুত ও কম খরচের রোবট প্রত্যাশা করি
- রোগ- বালাই নির্ণয়, ও পরিচর্যা

GOALS AND BENEFITS OF THE PROJECT:

GOALS OF THE PROJECT:

The main goal of this project is to develop a Smart Irrigation Rover for Adaptive Spot Watering that can move through a small agricultural area, measure soil moisture at different points, and apply water only to the dry spots. Instead of using a fixed soil moisture sensor or watering the whole area, the rover will identify moisture conditions at multiple soil points and perform spot-based irrigation.

Another goal is to classify soil conditions into different moisture zones such as Dry, Moderate, and Wet. Based on this classification, the rover will control the water pump for different durations so that very dry soil receives more water, moderately dry soil receives less water, and wet soil receives no water.

The project also aims to include post-irrigation verification, where the rover checks the soil moisture again after watering to confirm whether the irrigation was effective. In addition, IoT monitoring will be used to display soil moisture, temperature, humidity, zone status, and pump activity.

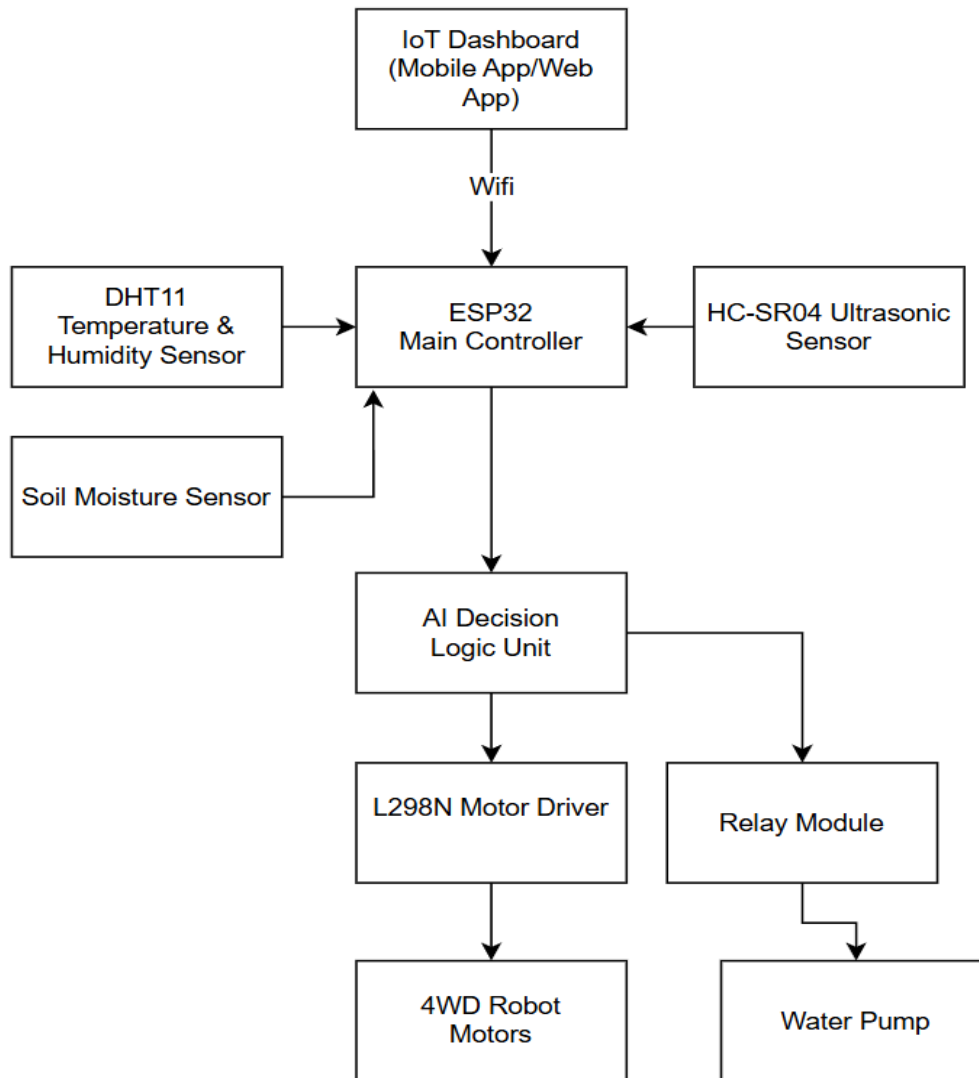
BENEFITS OF THE PROJECT:

This project can reduce unnecessary water usage by applying water only to the soil points that require irrigation. Since the rover checks multiple soil locations, it can identify dry spots more effectively than a fixed single-sensor irrigation system.

The adaptive spot watering method helps improve irrigation efficiency by controlling pump operation based on soil dryness level. The post-irrigation verification feature also helps confirm whether the watering action improved the soil moisture condition.

The system reduces manual checking of soil moisture and provides remote monitoring through IoT. Overall, the project demonstrates a low-cost smart irrigation solution that can support future research in precision irrigation, mobile soil monitoring, and autonomous agricultural robotics.

EXPERIMENTAL BLOCK DIAGRAM:



The proposed Smart Irrigation Rover for Adaptive Spot Watering is controlled by an ESP32 microcontroller, which acts as the central processing unit of the system. The soil moisture sensor, DHT11 sensor, and ultrasonic sensor continuously collect data about soil condition, environmental parameters, and obstacles. Based on the collected information, the system decides whether irrigation is required and controls the rover's movement and water pump accordingly. The rover can automatically move through the field, identify dry areas, and provide spot-based watering where needed. All important data, including soil moisture, temperature, humidity, and irrigation status, are transmitted to an IoT dashboard for real-time monitoring and analysis.

PROJECT TIMELINE (GANTT CHART):

Activities	June W1	June W2	June W3	June W4	July W1	July W2	July W3	July W4	August W1	August W2
Literature Review										
Survey & Interview										
Component Selection										
Hardware Development										
Software Development										
Testing & Integration										
Research Paper Writing										
Final Presentation										

REFERENCES:

- [1] K. Obaideen, B. A. A. Yousef, M. N. AlMallahi, Y. C. Tan, M. Mahmoud, H. Jaber, and M. Ramadan, "An overview of smart irrigation systems using IoT," *Energy Nexus*, vol. 7, 2022.
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- [3] V. Kumar S., C. D. Singh, K. V. R. Rao, M. Kumar, Y. A. Rajwade, B. Babu, and K. Singh, "Evaluation of IoT based smart drip irrigation and ETc based system for sweet corn," *Smart Agricultural Technology*, vol. 5, 2023.
- [4] M. M. Alboaneen et al., "IoT-enabled smart agriculture for improving water management: A smart irrigation control using embedded systems and Server-Sent Events," *Results in Engineering*, vol. 25, 2025.
- [5] C. Han, W. Wu, X. Luo, and J. Li, "Visual navigation and obstacle avoidance control for agricultural robots via LiDAR and camera," *Remote Sensing*, vol. 15, no. 22, 2023.
- [6] F. Soto-Meca, J. M. Molina-Martínez, Á. Soto-García et al., "Comparative analysis of soil moisture- and weather-based irrigation scheduling for drip-irrigated lettuce using low-cost IoT capacitive sensors," *Sensors*, vol. 25, no. 5, 2025.
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COMMENTS BY COURSE TEACHER:

COURSE TEACHER’S NAME
DATE

COURSE TEACHER’S SIGNATURE

GROUP MEMBERS

NAME: Akhi Ketu Chakma ID: 2210952 PROGRAM: CSC EMAIL: 2210952@iub.edu.bd Phone: 01894852404 Semester No: 11 th	NAME: Md. Jawad Abdullah ID: 2230325 PROGRAM: CSE EMAIL: 2230325@iub.edu.bd Phone: 01303127053 Semester No: 13 th
NAME: Shajalal Sojib ID: 2220117 PROGRAM: CSC EMAIL: 2220117@iub.edu.bd Phone: 01831145318 Semester No: 13 th	NAME: Abdul Gaffar Fahim ID: 2222752 PROGRAM: CSE EMAIL: 2222752@iub.edu.bd Phone: 01850824802 Semester No: 14 th
NAME: Ashibul Islam Ashif ID: 2230191 PROGRAM: CSE EMAIL: 2230191@iub.edu.bd Phone: 01630001746 Semester No: 13 th	
REMARKS (for OFFICE use only)	